

Homework 15 Solutions

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Problem 33.1

Let (S, P) be a sample space with $S = \{a, b, c, d\}$ and $P = \{(a, 0.1), (b, 0.2), (c, 0.3), (d, 0.4)\}$. Let X and Y be the random variables given in the table.

- a) The event " $X > 3$ " is $\{c, d\}$, and $P(X > 3) = P(c) + P(d) = 0.7$.
- b) The event " Y is odd" is $\{a, b\}$, and $P(Y \text{ is odd}) = P(a) + P(b) = 0.3$.
- c) The event " $X > Y$ " is $\{a\}$, and $P(X > Y) = P(a) = 0.1$.
- d) The event " $X = Y$ " is $\{b\}$, and $P(X = Y) = P(b) = 0.2$.
- e) $P(X = 1 \text{ and } Y = -1)$ is 0.1, $P(X = 3 \text{ and } Y = 3)$ is 0.2, $P(X = 5 \text{ and } Y = 6)$ is 0.3, $P(X = 8 \text{ and } Y = 10)$ is 0.4, and for all other combinations of integers (n, m) , then $P(X = n \text{ and } Y = m)$ is 0.
- f) X and Y are NOT independent since knowing X gives you information about Y . For example, if you know X is 1, then you know Y is -1.
- g) Let Z be the random variable $Z = X + Y$. Then, $P(Z = 0)$ is 0.1, $P(Z = 6)$ is 0.2, $P(Z = 11)$ is 0.3, $P(Z = 18)$ is 0.4, and then for any other integer n , $P(Z = n)$ is 0.

Problem 33.2

Let (S, P) be a sample space with $S = \{1, 2, 3, \dots, 10\}$, $P(s) = \frac{1}{10}$, and define the random variables X and Y by $X(s) = 2s$ and $Y(s) = s^2$, $\forall s \in S$.

- a) $P(X < 10) = P(X = 2) + P(X = 4) + P(X = 8) = 0.4$.
- b) $P(Y < 10) = 0.3$.
- c) $(X + Y)(s) = 2s + s^2$.
- d) $P(X + Y < 10) = 0.2$.
- e) $P(X > Y) = 0.1$.
- f) $P(X = Y) = 0.1$.
- g) X and Y are NOT independent since knowing X gives you information about Y . For example, if you know X is 6, then you know Y is 9.

Problem 33.9

The random variables X_1 and X_2 are NOT random variables. Observe that the values of X_1 and X_2 are always within 1 of each other (if $X_1 = 10$, then X_2 is either 9 or 11). Therefore knowing information about X_1 gives you information about X_2 .

Problem 34.1

$$\begin{aligned}
E(X) &= 1 \cdot P(X = 1) + 3 \cdot P(X = 3) + 5 \cdot P(X = 5) + 8 \cdot P(X = 8) = 1 \cdot 0.1 + 3 \cdot 0.2 + 5 \cdot 0.3 + 8 \cdot 0.4 = 5.4 \\
E(Y) &= -1 \cdot P(Y = -1) + 3 \cdot P(Y = 3) + 6 \cdot P(Y = 6) + 10 \cdot P(Y = 10) = -1 \cdot 0.1 + 3 \cdot 0.2 + 6 \cdot 0.3 + 10 \cdot 0.4 = 6.3 \\
E(Z) &= 0 \cdot P(Z = 0) + 6 \cdot P(Z = 6) + 11 \cdot P(Z = 11) + 18 \cdot P(Z = 18) = 0 \cdot 0.1 + 6 \cdot 0.2 + 11 \cdot 0.3 + 18 \cdot 0.4 = 11.3
\end{aligned}$$

Problem 34.2

Let (S, P) be a sample space with $S = \{a, b, c\}$, $P(s) = \frac{1}{3}$ for all $s \in S$, and let X , Y , and Z be random variables given by $X = \{(a, 1), (b, 2), (c, 10)\}$, $Y = \{(a, -1), (b, -1), (c, 2)\}$, and $Z = X + Y$.

a) $E(X) = 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + 10 \cdot \frac{1}{3} = \frac{13}{3}.$

b) $E(Y) = -1 \cdot \frac{1}{3} + -1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} = \frac{0}{3}.$

c) $E(Z) = 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 12 \cdot \frac{1}{3} = \frac{13}{3}.$

Problem 34.4

Let (S, P) be a sample space with $S = \{1, 2, 3, 4, 5, 6\}$, $P(s) = \frac{1}{6}$, and let the random variable $X(s) = s^2$ for all $s \in S$. Then $E(X) = 1^2 \cdot \frac{1}{6} + 2^2 \cdot \frac{1}{6} + 3^2 \cdot \frac{1}{6} + 4^2 \cdot \frac{1}{6} + 5^2 \cdot \frac{1}{6} + 6^2 \cdot \frac{1}{6} = (1 + 4 + 9 + 16 + 25 + 36) \cdot \frac{1}{6} = \frac{91}{6}$, so approximately \$15.17.